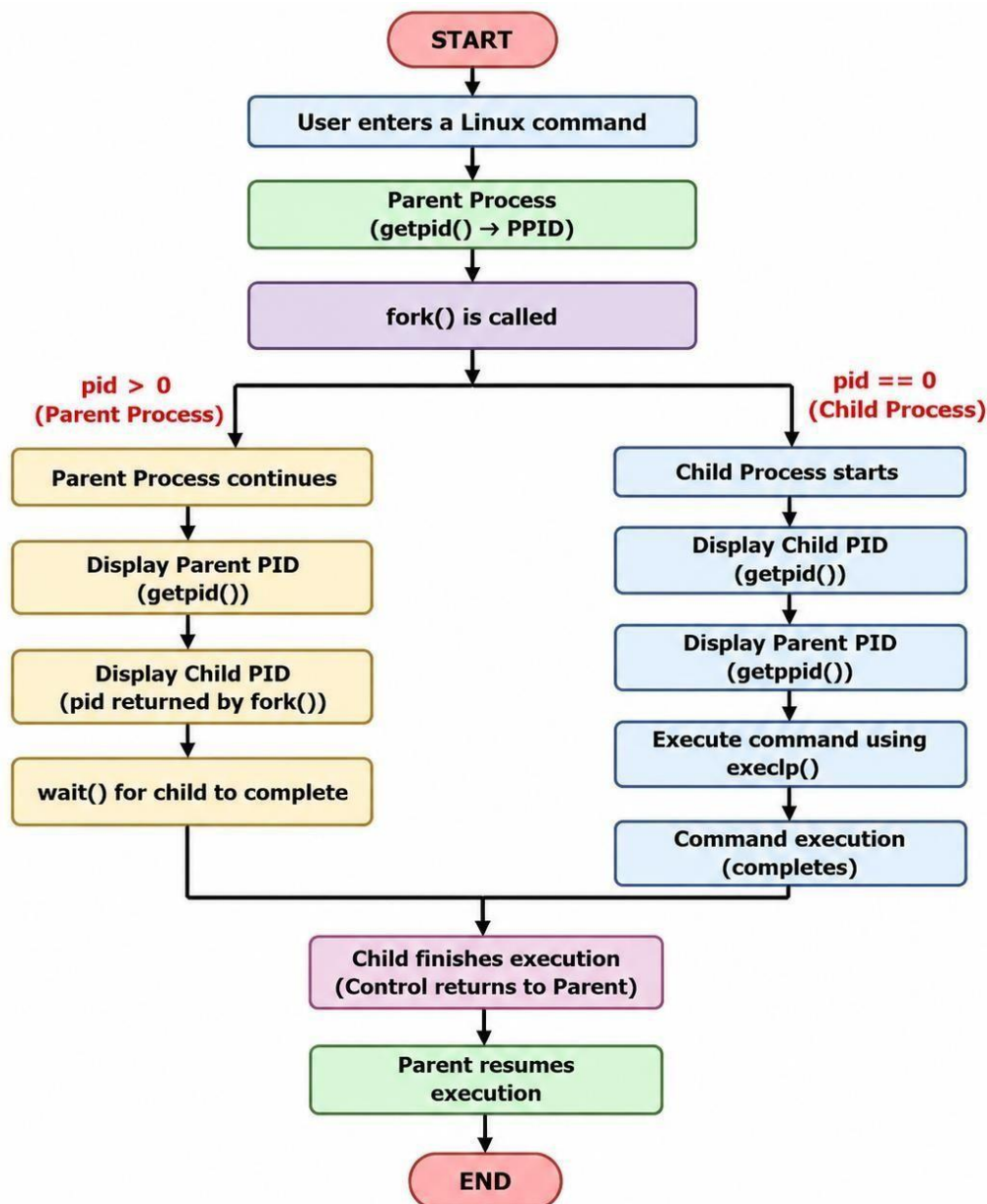


Practical Week -1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait()`.
5. Display the Process ID (PID) of both parent and child processes.

- Write a C code to create a process using `fork()` system call.
- Read the user input as shell command.
- Use a system call `exec()` to execute the shell command.
- Use a `wait()` system call for waiting a parent process.
- Display the PIDs of parent and child processes.



Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

Executing command: ls

CE1 FIRST.c File2.c Proces_States.c command_executor
filecopy.c hello.c

process_example session_4_a session_4_b.c

CE1.c File1.c First.c Process_States.c command_executor.c
fork_example.c.save my_shell process_example.c session_4_a.c

FIRST File1.txt Proces_States R1 filecopy hello
os-lab process_example.cyes session_4_b

Child process completed.

Sol:

```
GNU nano 8.7.1      command_executor.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }

    else if (pid == 0)
    {
        // Child process

        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("Executing command: %s\n", command);

        execlp(command, command, NULL);
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify

```
}

else if (pid == 0)
{
    // Child process

    printf("Child Process\n");
    printf("Child PID : %d\n", getpid());
    printf("Parent PID : %d\n", getppid());

    printf("Executing command: %s\n", command);

    execlp(command, command, NULL);

    perror("exec failed");
    exit(1);
}

else
{
    // Parent process

    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}
```

^G Help
^X Exit

^O Write Out
^R Read File

^F Where Is
^ Replace

^K Cut
^U Paste

^T Execute
^J Justify

```

sirichandana@siri: ~
File1.txt      ecec_demo.c  fork_exec.c  task2.c
File2.txt      exec_demo    namedirectory task3
OSSP           exec_demo.c  skill4       task3.c
OS_LAB         filename     task1
command_executor.c.save filename.c    task1.c
ecec_demo      fork_exec    task2
sirichandana@siri:~$ nano command_executor.c
sirichandana@siri:~$ gcc command_executor.c -o command_executor
sirichandana@siri:~$ ./command_executor
Enter a Linux command: ls
Parent Process
Parent PID : 550
Child Process
Child PID : 551
Parent PID : 550
Executing command: ls
File1.txt      ecec_demo.c  skill4
File2.txt      exec_demo    task1
OSSP           exec_demo.c  task1.c
OS_LAB         filename     task2
command_executor filename.c    task2.c
command_executor.c fork_exec    task3
command_executor.c.save fork_exec.c  task3.c
ecec_demo      namedirectory
Child process completed.
sirichandana@siri:~$ ./command_executor
Enter a Linux command: pwd
Parent Process
Parent PID : 617
Child Process
Child PID : 620
Parent PID : 617
Executing command: pwd
/home/sirichandana
Child process completed.
sirichandana@siri:~$

```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Sol:

Command

uname

uname -a

What it

shows

Linux/system
information

Detailed
kernel/system

lscpu

ps aux

top

em
information
CPU

architectur
e and

processor information Current ps
processes

Detailed
process
information

Real-time
process

and
resource
usage **What**
it

Command shows

```
sirichandana@siri:~$ uname  
Linux  
sirichandana@siri:~$ uname -a  
Linux siri 6.18.33.2-microsoft-standard-WSL2 #1  
REEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux  
sirichandana@siri:~$ lscpu  
  
Architecture:                x86_64  
CPU op-mode(s):              32-bit, 64-bit  
Address sizes:               39 bits physical, 48 bits virtual  
Byte Order:                  Little Endian  
CPU(s):                      12  
On-line CPU(s) list:        0-11  
Vendor ID:                   GenuineIntel  
Model name:                  13th Gen Intel(R) Core(TM) i  
CPU family:                   6  
Model:                       186  
Thread(s) per core:          2  
Core(s) per socket:          6  
Socket(s):                   1  
Stepping:                    3  
BogoMIPS:                    4991.99  
Flags:                        fpu vme de pse tsc msr pae m  
apic sep mtrr pge mca cmov p  
6 clflush mmx fxsr sse sse2  
syscall nx pdpe1gb rdtscp lm  
t_tsc rep_good nopl xtopolog  
eliable nonstop_tsc cpuid ts  
_freq pni pclmulqdq vmx ssse  
xl6 pcid sse4_1 sse4_2 x2api  
popcnt tsc_deadline_timer a  
e avx f16c rdrand hypervisor  
m abm 3dnowprefetch ssbd ibr  
stibp ibrs_enhanced tpr_shad  
vpid ept_ad fsgsbase tsc_adj  
1 avx2 smep bmi2 erms invpci  
saveopt xsaver xgetbv1 xsave
```

```

sirichandana@siri:~$ ps
  PID TTY          TIME CMD
  319 pts/0    00:00:00 bash
  733 pts/0    00:00:00 ps

sirichandana@siri:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TI
ME COMMAND
root           1  0.3  0.3  24328 15372 ?        Ss   05:20   0:
03 /sbin/init
o0 /init       2  0.0  0.0   3180  2208 hvc0     Sl+  05:20   0:
root          7  0.0  0.0   3180  2208 ?        Sl+  05:20   0:
00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-f
root         53  0.0  0.4  50396 16804 ?        Sss  05:20   0:
00 /usr/lib/systemd/systemd-journald
systemd+     70  0.0  0.3  22416 14468 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-resolved
root         92  0.1  0.3  35140 12256 ?        Ss   05:20   0:
01 /usr/lib/systemd/systemd-udevd
root        153  0.0  0.0   2888  1948 ?        Ss   05:20   0:
00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n --F 1
root        154  0.0  0.0   4472  3100 ?        Ss   05:20   0:
00 /usr/sbin/cron -f -P
message+    155  0.0  0.1   8820  5316 ?        Ss   05:20   0:
00 @dbus-daemon --system --address=systemd: --nofork --nopidfile
root        159  0.0  0.7  44096 29712 ?        Ss   05:20   0:
00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-t
root        168  0.0  0.2  18436  9388 ?        Ss   05:20   0:
00 /usr/lib/systemd/systemd-logind
syslog      175  0.0  0.1  220548 5568 ?        Ssl  05:20   0:
00 /usr/sbin/rsyslogd -n -iNONE
root        216  0.0  0.0   5492  2660 tty1     Ss+  05:20   0:
00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:
root        225  0.0  0.8 123460 32772 ?        Ssl  05:20   0:
00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-up
_chrony     226  0.0  0.2   2388 11084 ?        S    05:20   0:
00 /usr/sbin/chronyd -n -F 1 -x

```

```

sirichandana@siri:~$ top
top - 05:43:42 up 2:51, 1 user, load average: 0.09, 0.04, 0.02
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si
MiB Mem : 3758.6 total, 3048.4 free, 473.1 used, 312.9 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3285.5 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM
   1 root        20   0   24328 15572 11568 S    0.0   0.4
   2 root        20   0   3180   2208 2072 S    0.0   0.1
   7 root        20   0   3180   2048 1940 S    0.0   0.1
  53 root        19  -1   50396 16804 15500 S    0.0   0.4
  70 systemd+   20   0   22416 14468 11972 S    0.0   0.4
  92 root        20   0   35140 12256 9184 S    0.0   0.3
 153 root        20   0   2888   1948 1820 S    0.0   0.1
 154 root        20   0   4472   3100 2852 S    0.0   0.1
 155 message+   20   0   8820   5316 4636 S    0.0   0.1
 159 root        20   0   44096 29712 14576 S    0.0   0.8
 168 root        20   0   18436  9388 8144 S    0.0   0.2
 175 syslog     20   0  220548 5568 4648 S    0.0   0.1
 216 root        20   0   5492   2660 2476 S    0.0   0.1
 225 root        20   0 123460 32772 18164 S    0.0   0.9
 226 _chrony     20   0  23588 11084 7128 S    0.0   0.3
 237 _chrony     20   0  12096  2452 1808 S    0.0   0.1
 316 root        20   0   3184   1112  980 S    0.0   0.0
 319 root        20   0   3200   1248 1108 S    0.0   0.0
 319 sirichandana 20   0   6268  5532 3864 S    0.0   0.1
 322 root        20   0   8648  5544 4700 S    0.0   0.1
 385 sirichandana 20   0  22340 13604 10976 S    0.0   0.4
 387 sirichandana 20   0  22920  4092 2088 S    0.0   0.1
 412 sirichandana 20   0   6136  5424 3864 S    0.0   0.1
 755 sirichandana 20   0   8164  6064 3948 R    0.0   0.2

```